Head2Wrist: Wrist-Camera Robot Policies from Human Head-Camera Demonstrations without a Scene Scan

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Abstract: Human head-camera demonstrations make up most large open manipulation datasets, but a robot trained on them meets two gaps: a moving human head rarely reproduces the robot's head view, and the arm in the images is human; inpainting a robot arm over it keeps the first. We instead train on the robot's wrist camera, whose view rides on the gripper, and generate it from a single human head-camera demonstration (posed head video and hand tracking), with no separate scene scan and no teleoperation. We reconstruct the scene from the pre-grasp head frame and nearby posed frames, plan a robot-executable, hand-aligned trajectory checked in a physics twin, and render the wrist views with layers matched to the robot's camera (dynamic object, gripper, robot body, cast shadows). In closed loop on the real FFW-SG2, a policy trained only on views generated from three Meta Quest 3 demonstrations (a cube into a bowl) succeeds in 18 of 20 trials from cube starts within 3 cm and 30° of a demonstrated one, where replaying the planned path open loop succeeds in 5. In a matched training-image comparison in Isaac Lab twins of six EgoDex scenes, with the generator given a clean twin head render and the object's exact geometry and start pose, ACT policies trained only on generated views show no detectable difference from true-view policies on three development scenes (all 9 paired $p \geq 0.12$); on two held-out scenes they succeed less often in range (47/60 vs 58/60) and far less outside it (4/40 vs 27/40), and on a 1.4 cm-thin phone they succeed in 3 of 30 in-range trials (true view: 15).

Keywords: egocentric demonstrations, wrist-view synthesis, imitation learning

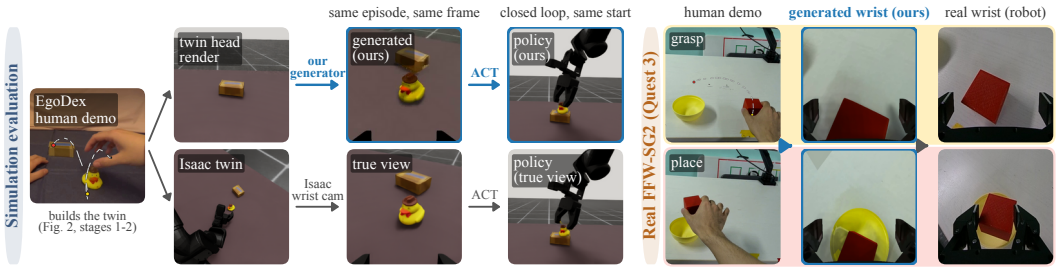


Figure 1: **Head2Wrist** evaluation. Left: an EgoDex demonstration (stack/48; dashed: hand path) becomes an Isaac Lab twin whose head render feeds our generator; both ACT policies are tested from one start (right: their rollouts). Right, at grasp and place: a Quest 3 demonstration (a cube into a bowl), our generated wrist views and the robot's real ones (plan replay).

1 Introduction

Most human manipulation data, from VR headsets [1] to large open egocentric datasets [2], is head-camera video, yet wrist cameras make manipulation policies precise near contact (UMI [3] mounts

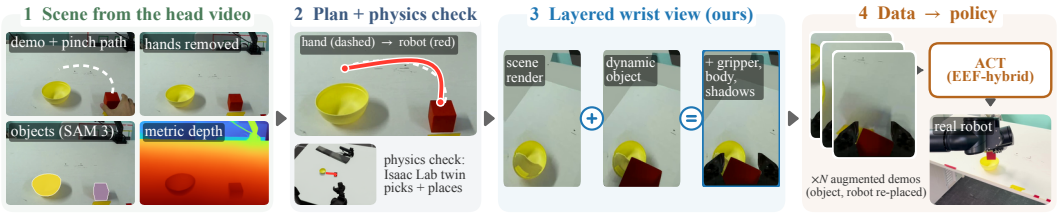


Figure 2: Method on one real Quest 3 demonstration (a cube into a bowl). (1) The pre-grasp head frame and nearby posed frames become a metric scene; (2) a hand-aligned robot path (red; the human's dashed) is checked in a physics twin; (3) wrist views are rendered along it with the robot's layers; (4) demo augmentation: the plan is redone for new object starts and robot stances, giving training episodes; real robot: photo of this plan's replay.

one on a handheld gripper). Policies trained on it see the human only through the head view, which a moving human rarely holds at the robot's pose, and see a human arm that is absent at test time. They add wrist views from robot data [4, 5], extract 3D states [6, 7], or replace the human with a rendered robot in the same head view [8, 9], which leaves the viewpoint gap. WristWorld [10] and in-hand view synthesis [11] learn it from paired robot views, and WARPED [12] renders wrist views of egocentric demonstrations from a Gaussian-splat reconstruction of a separate workspace scan.

Head2Wrist instead trains on the robot's wrist view, generated from the human's posed head-camera video and hand track with no separate scene scan and no teleoperation (Fig. 1). Our contributions: (1) a scan-free pipeline from one head-camera demonstration to robot wrist-camera training views along a hand-aligned, physics-checked robot trajectory, with a Quest 3 converter and an FFW-SG2 interface; (2) a closed-loop real-robot test (Fig. 1, right) in which a policy trained only on generated views succeeds in 18 of 20 trials (open-loop replay: 5); (3) a matched training-image comparison in sim (Fig. 1, left): given a clean twin head render and the object's exact geometry and start pose, generated-view policies show no detectable difference from true-view ones on three of four development scenes but a gap on two held-out scenes, largest out of range; and a recipe that makes such data usable (end-effector hybrid actions, stance randomization, strong augmentation).

2 Method

Scene from the head video (Fig. 2). From the head frame just before the grasp we remove the hand by inpainting and estimate depth with Depth Anything 3 [13] on nearby head frames, given their recorded camera poses; one scale factor matches its table plane to the metric hand track. We fit the table, segment the objects (SAM 3 [14]) and lift the grasped object to a mesh (SAM 3D [15], depth-refined); the scene proxy (depth mesh, objects, table) extends the table past the head's view.

Robot plan. The robot is the ROBOTIS FFW-SG2, a bimanual robot with an Intel RealSense D405 on each wrist. Its tool center point (TCP) path follows the demonstration: an overview pose that sees the objects, a descent to a grasp aligned with the human pinch and palm, lift, transport, set-down at the human's release point, and retreat; the D405 pose follows from the URDF. As on the real robot, the base and lift are placed once: a grid search over base position, heading and lift picks the stance with the largest joint-limit margin from which arm-only IK reaches the whole path; for the hand-placed real robot it must also tolerate placement errors up to 5 cm and 10° (feasible random placements: 0/16 without it, 15/16 with it). An Isaac Lab [16] twin replays the planned motion with physics and tries grasp variants until one picks and places, else rejects the demonstration.

Wrist-view generator. Along the planned D405 path we render the proxy (each surface point takes the head-frame pixel it projects to if unoccluded, else an outpainted table map) and composite four layers matched to the target camera: the dynamic object, the grasped object's mesh moving with the gripper; the URDF gripper at the planned opening (closed to the object's planned width), textured from that camera's own frames; the robot's arm and body, hidden behind the table; and the shadows of these from a randomized light (tilt 0–35°, darkening 0.3–0.6).

Policy. ACT [17] on the active arm's 424×240 wrist image, with an end-effector (EEF) hybrid representation: the state is the absolute TCP pose in the base frame plus the jaw width; the action chunk is TCP poses relative to the current one plus the absolute width, executed through IK. Training randomizes the stance (forward −5 to +2 cm, sideways ±5 cm, yaw ±10°), with strong augmentation (affine, colour, blur, noise).

3 Experiments

Real robot. Of 4 Quest 3 demonstrations (a cube into a bowl), 3 passed the physics check; replanning them for object starts within ±8 cm and ±30° and randomized stances gave 252 training episodes. On 20 test starts inside that range, within 3 cm and 30° of one demonstration's start, the policy trained only on generated views succeeded (cube in the bowl) in 18 (both failures: released outside it); replaying its planned path open loop, which cannot react to the cube, succeeded in 5 (Fig. 3, right). This tests closed-loop use; the sim test isolates the images. Along a replayed path, the generated views match the real D405 frames to SSIM [18] 0.64 at a median misalignment of 5.6 px (sim: 0.84, 0.3 px; Fig. 1, right). Quest 3 (EgoKit [1]) recordings are converted to the EgoDex format, and a ROS 2 bridge steps the policy per reached pose.

Simulation protocol. Scenes are EgoDex test episodes, named task/episode (pick-place: basic_pick_place). Of 8 screened scenes, 5 pass the physics check: stack/18, stack/48, pick-place/25 and pick-place/1 served development and pick-place/73 was held out, joined by pick-place/67, the first of a later screen to pass unaided. Each gets an Isaac Lab twin. A scripted expert executes re-planned paths for randomized object starts and robot stances, recorded by the true sim D405 (225–630 demonstrations per scene). The generated-view dataset replaces every wrist frame of the same trajectories with the generator's output, made from one clean head-view render of the twin (standing in for the human's head frame) and the object's mesh and start pose. Policies are evaluated closed loop on the *true* camera: 30 (50 on stack/18) in-range object starts (training: ±8 cm, ±30°), 20 (40) starts up to 12 cm with an axis outside that range, and 24 in-range starts with the base moved by up to 5 cm and 10°. An episode succeeds only if the object is lifted, the gripper is open at the end, and the object ends within 3 cm (XY) and 2 cm (height) of where the expert leaves it.

No detectable gap on the development scenes, a gap on the held-out ones (Fig. 3). On stack/18, stack/48 and pick-place/25 none of the nine paired comparisons differs significantly (stack/48 out of range: both far below the expert, 5/20 and 6/20 vs 16/20); on pick-place/73 and pick-place/67, with the generator and recipe frozen, the generated-view policy is lower in all six sets: pooled 47/60 vs 58/60 in range, 4/40 vs 27/40 outside it, 25/48 vs 35/48 moved base (exact $p = 0.003$, $< 10^{-4}$, 0.02).

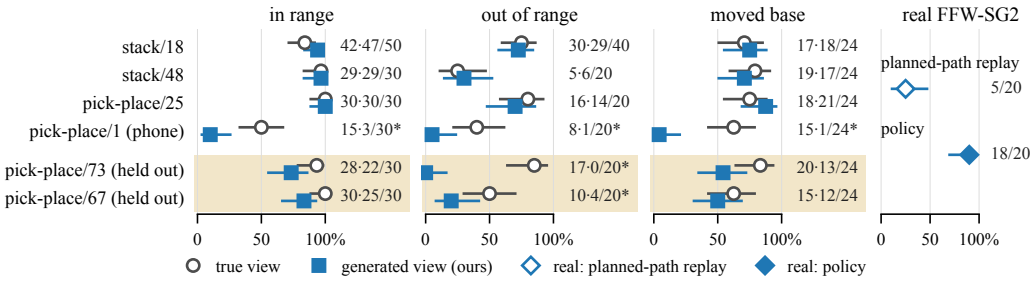


Figure 3: Task success with 95% Wilson intervals; beside each pair: true · generated / episodes, ∗: paired exact McNemar $p < 0.05$. Simulation (left): the two ACT policies differ only in their training wrist images (same trajectories, states, actions) and are tested on the true D405 from paired starts; stack/18 out of range pools 16 + 24 starts; shaded: held out. Real FFW-SG2 (right, cube into bowl): open-loop plan replays and generated-view policy trials on the same 20 cube starts.

Table 1: Ablations (closed loop, true camera; in / out of range, moved base). Left: generated views, stack/18 unless noted; EEF absolute: absolute TCP-pose actions; ‡per-episode error of the generator's object start pose and size (scale: median 2.5 / 4.9%); †constant grey wrist image in training and test, stack/48 + pick-place/25, stopped early. Right: bundled generator versions (successes; base: scene, object, gripper; colour: colour-matched table extension, target-camera gripper texture; seamless: seamless extension; body: robot body, shadows).

setting	in	out	moved
joint-space ACT, fixed stance	22/50	4/40	0/24
EEF absolute, fixed stance	45/50	22/40	2/24
EEF hybrid, fixed stance	45/50	27/40	7/24
+ stance randomization	21/50	20/40	7/24
+ strong aug. (base generator)	46/50	33/40	17/24
+ full generator	47/50	29/40	18/24
true views (same recipe)	42/50	30/40	17/24
generated, exact object (stack/48)	29/30	6/20	17/24
obj. error‡, median 1.2 cm / 3°	25/30	5/20	15/24
obj. error‡, median 3.0 cm / 6.5°	18/30	2/20	12/24
constant image†	0/35	0/33	1/34

	stack/48			pick-place/25		
gen.	in	out	moved	in	out	moved
base	3	0	3	6	3	2
+ colour	25	5	14	4	2	3
+ seamless	–	–	–	27	12	12
+ body (full)	29	6	17	30	14	21
trials	30	20	24	30	20	24

A second training seed on stack/48 and pick-place/25 also yields no significant paired difference (all six $p \geq 0.25$; between seeds all $p \geq 0.12$). On the 1.4 cm-thin phone (pick-place/1), generated versus true views yield 3/30 versus 15/30 in-range successes and 18 versus 10 rollouts with no lift.

Representation and training choices matter (Table 1, left). Joint-space ACT reaches 44% / 10% (in / out of range) on generated views, end-effector states (absolute or hybrid) 90% / 55–68%; randomizing the stance drops the hybrid policy to 42% until strong augmentation recovers 92% / 82% / 71%. **Object accuracy matters.** Perturbing the generator's object start pose and size per episode lowers the generated-view policy on stack/48 from 29/30 in range to 25/30 at a median error of 1.2 cm and 3° ($p = 0.22$ vs the exact-object policy, 0.12 vs the true view). At 3.0 cm and 6.5° it falls to 18/30 ($p < 0.001$ against both). The early-stopped constant-image control succeeds in 1 of 102 episodes (Table 1, left). Bundled generator revisions, chosen on the development scenes by inspection and teacher-forced image-region swaps, raised stack/48 in-range success from 3/30 (base layers) to 29/30 (full; Table 1, right); on stack/18 out of range it went from 33/40 to 29/40.

Video and world models as the generator. The generator must follow a close-range camera and keep the scene. In pilots, Lyra 2.0 [19], from the hand-removed head frame, drifted from the commanded camera near the table and GEN3C [20] dropped objects once the wrist camera magnified the head frame; over our scene proxy, Cosmos 3 [21] repainted the scene and VACE [22] inserted reference objects. Cosmos-Transfer2.5 [23] keeps the scene, but refining our views (27 GPU-min per episode) gave no gain on stack/48 at 20 demonstrations (refined vs ours: 21/30 vs 20/30 in range, 12/24 vs 13/24 moved base).

4 Limitations and conclusion

In sim the generator gets the object's geometry and start pose from the twin (real: head frame, SAM 3D), and synthetic object errors lower success. The four main scenes served development, and generator revisions were chosen on them, which may partly explain the gap on the held-out ones; a thin phone remains a failure case. No-difference results are not equivalence tests. Two main comparisons have a second seed; bundled generator revisions hide single-change effects. Real trials cover one task near its demonstrated start and, against open-loop replay, show closed-loop use, not an edge over other training data.

Head2Wrist turns one head-camera demonstration into wrist-camera training views without a scene scan: a policy trained only on them succeeded in 18 of 20 real trials; in sim they show no detectable gap on development scenes but fall short out of range on held-out ones (4/40 vs 27/40). Next, one world model should learn this whole process from these layered views and checked paths.

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